

I used non dimensional data given at the back of our text book (Page 416)

Flight stability and automatic control

By Robert C. Nelson

Q1.

A=

-0.0200	0.0159	0	-32.2000
-0.0427	-0.4035	869.3942	0
-0.0001	-0.0018	-0.5400	0
0	0	1.0000	0

B=

0.7809	9.6600
-18.5977	0
-1.2170	0
0	0

Q2.

2.1.

Short Period Mode Eigenvalues:

-0.4722 + 1.2614i
-0.4722 - 1.2614i

Long Period Mode Eigenvalues:

-0.0096 + 0.0300i
-0.0096 - 0.0300i

Short Period Mode Eigenvectors:

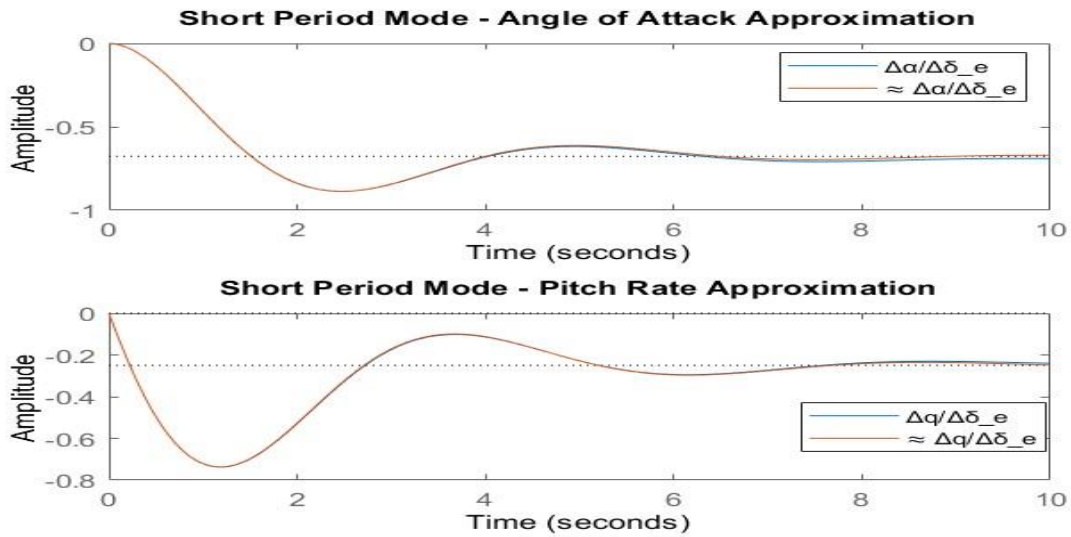
0.0117 + 0.0095i	0.0117 - 0.0095i
0.9999 + 0.0000i	0.9999 + 0.0000i
-0.0001 + 0.0015i	-0.0001 - 0.0015i
0.0010 - 0.0003i	0.0010 + 0.0003i

Long Period Mode Eigenvectors:

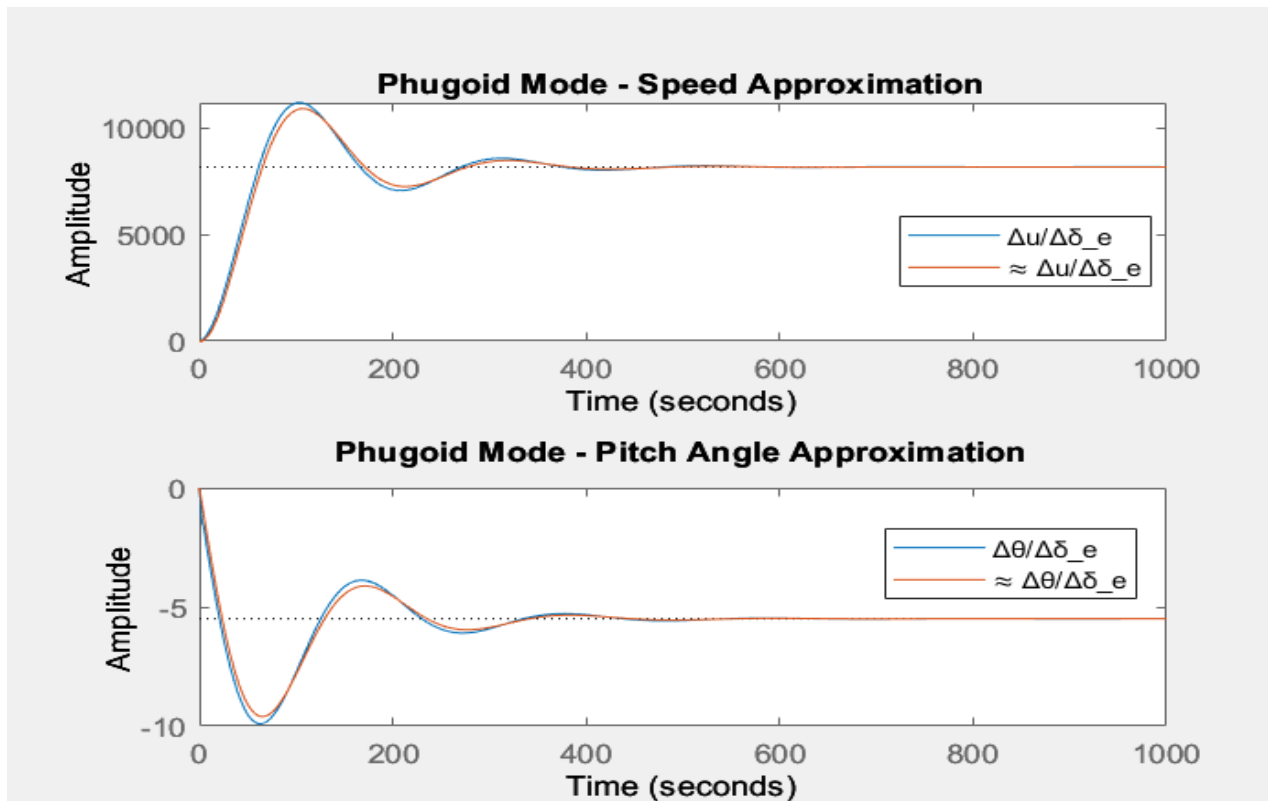
0.9992 + 0.0000i	0.9992 + 0.0000i
-0.0393 - 0.0001i	-0.0393 + 0.0001i
0.0000 - 0.0000i	0.0000 + 0.0000i
-0.0003 - 0.0009i	-0.0003 + 0.0009i

2.2.

Short Period approximation vs The exact responses:



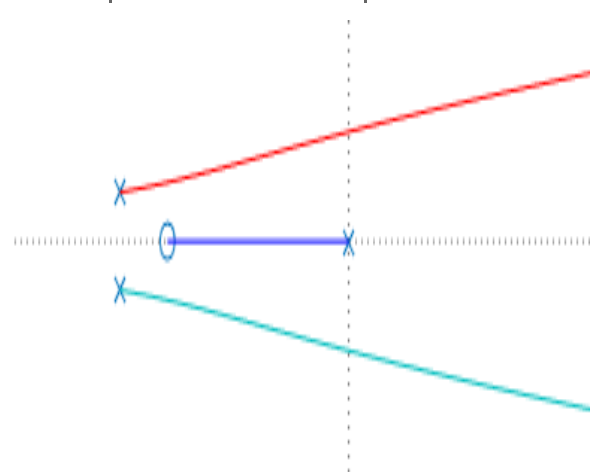
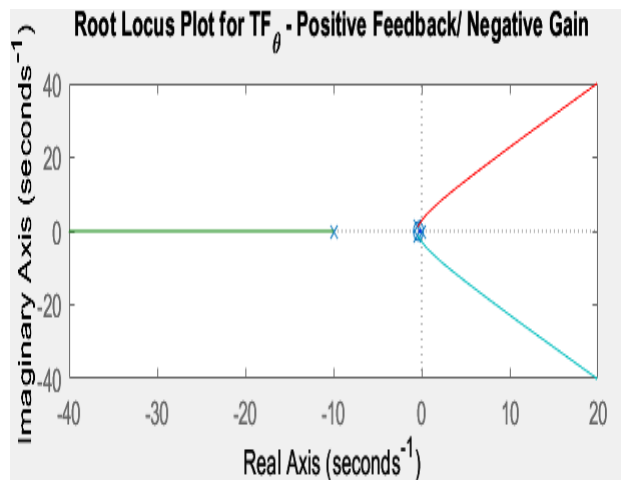
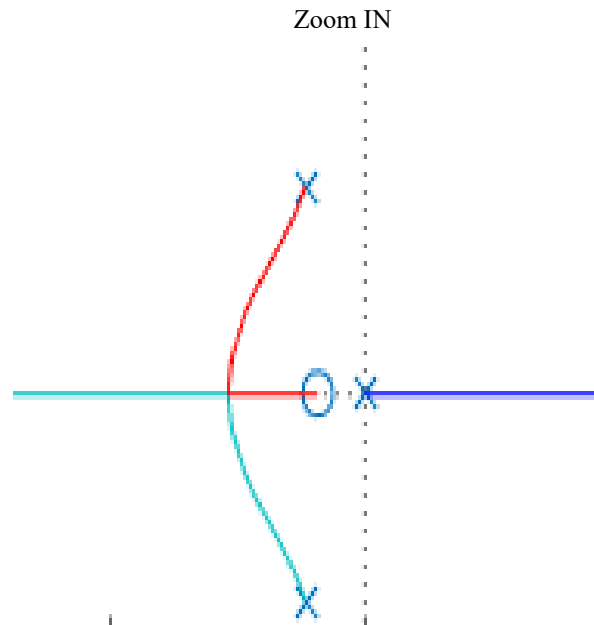
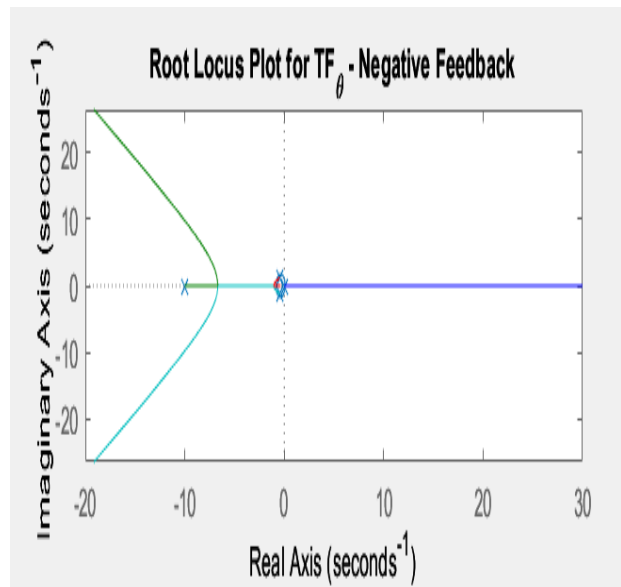
Long Period(Phugoid) approximation vs The exact responses:



Q3.

3.1 Pitch HOLD

For P controller:



As it is known, the dynamics of the pitch angle is both in the short period and phugoid mode. The above root locus is drawn by using the short period approximation. Two important things to observe:

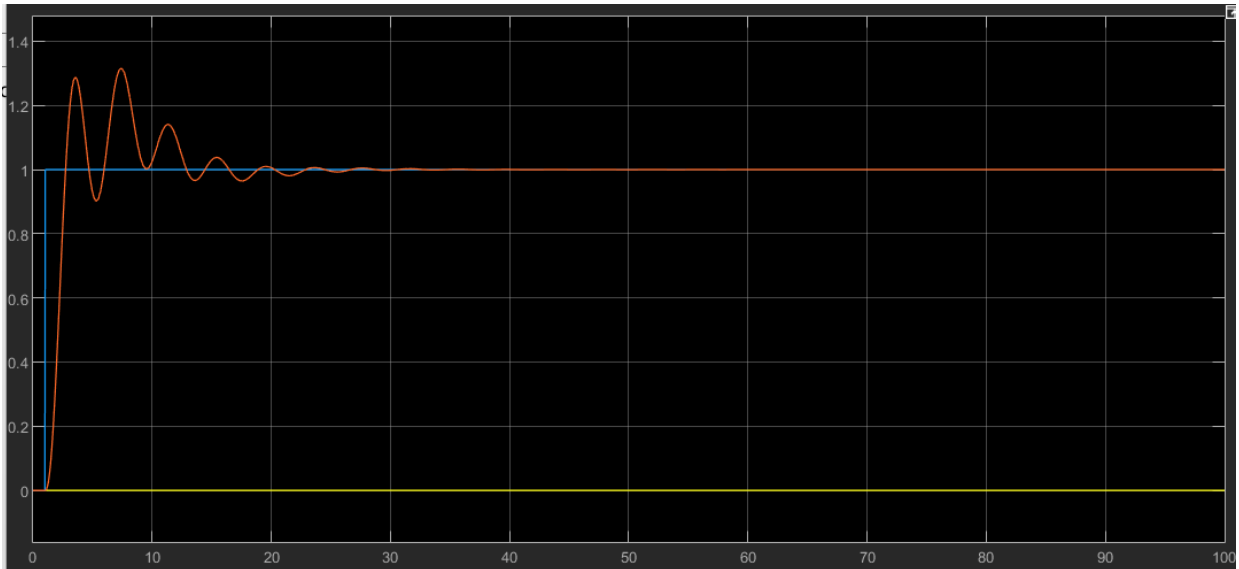
- The open loop has integrator in itself, so the steady error must be zero.
- For positive gain and negative feedback, the dynamics is unstable.

- Even for the negative gain and simple P controller, the gain shouldn't be greater than 5.

PI controller:

We have two degree of freedom to choose the parameters. Since there is no transient or steady state requirements for this problem, I chose the parameters in a way that the settling time is smaller.

PI:

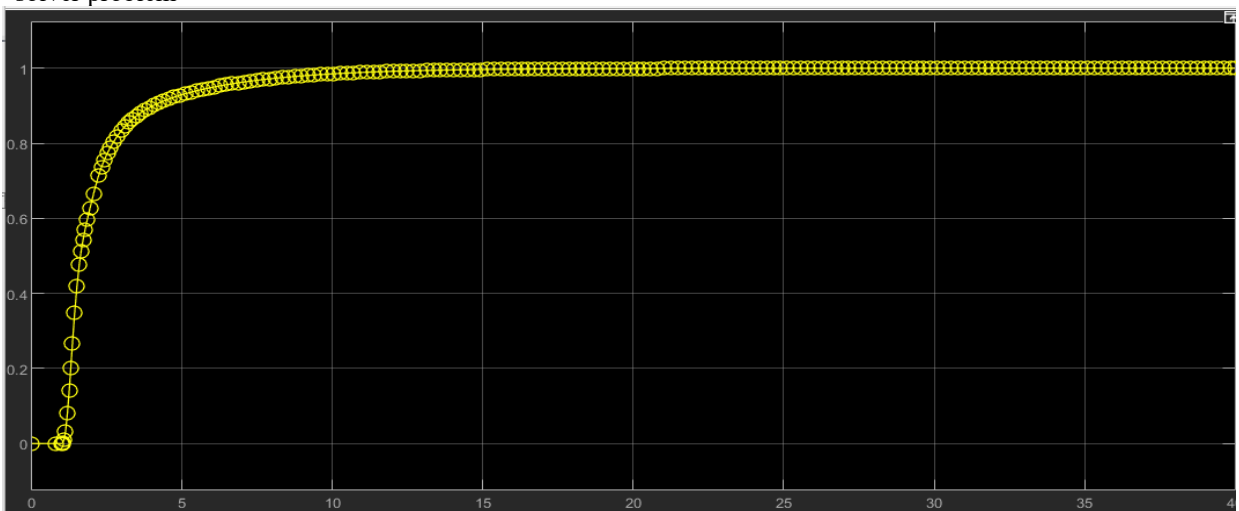


PD controller: We still have two degree of freedom to choose the parameters. Here are what I expect:

- Still zero steady state error. This is because of the integrator in the plant itself.
- More damping, so we can achieve less overshoot. This is because derivative adds damping to the system.

PD:

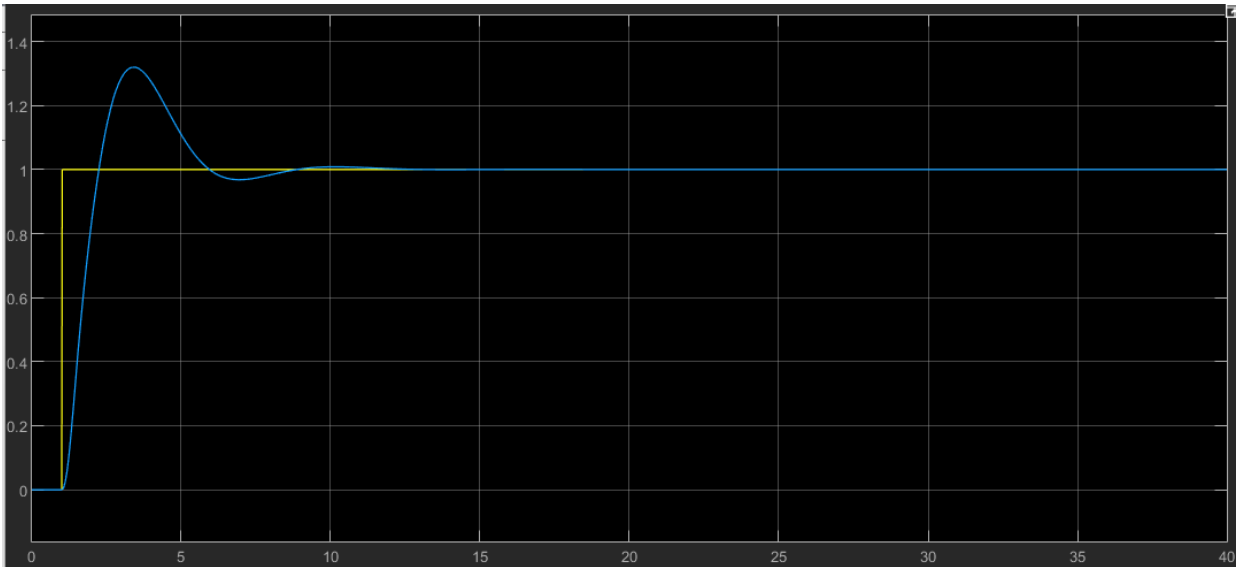
*solver problem



*Had the aircraft dynamics had no integrator, the PD can't achieve zero steady state error. Therefore, PI is might be better for that case!!

PID controller: Three degree of freedom to choose parameters. I can say this is the best because it can achieve zero steady state error for no integrator plant and also has pretty enough damping to avoid overshoot.

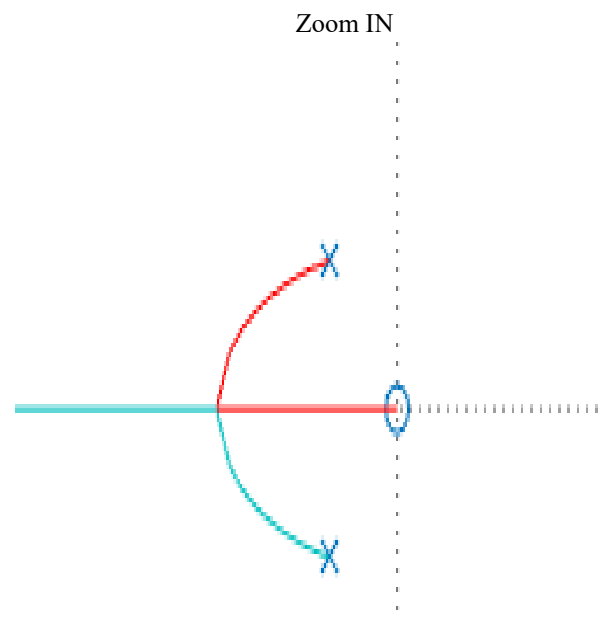
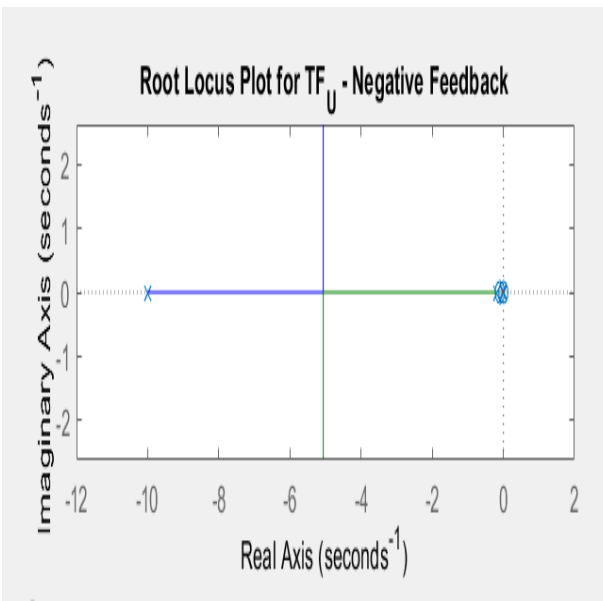
PID:

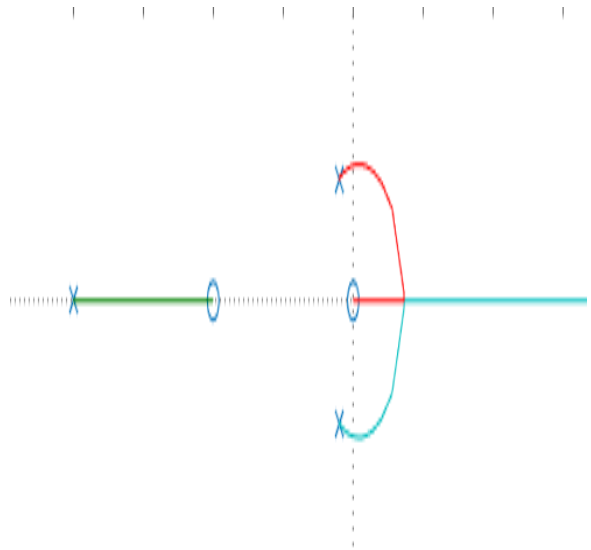
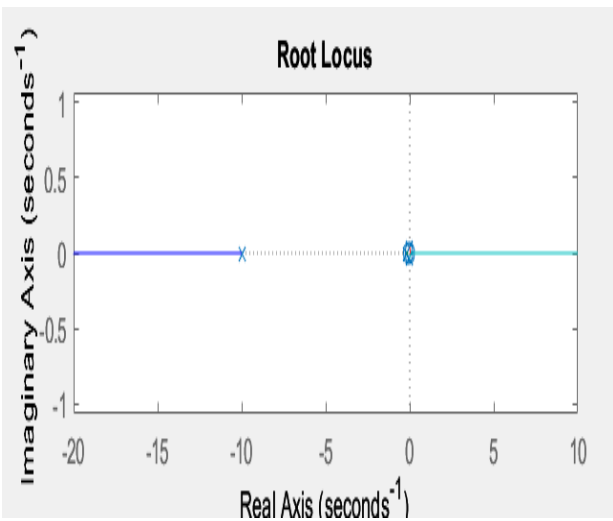


Please note that for each of the above controllers, I have analyzed using root locus and chose reasonable parameters.

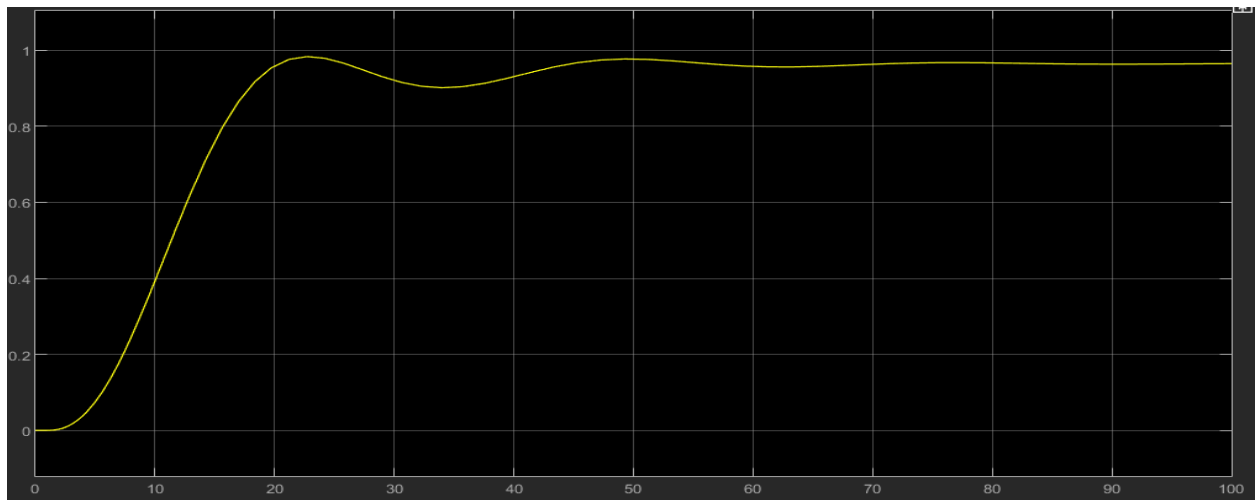
3.1 Speed HOLD

For P controller:

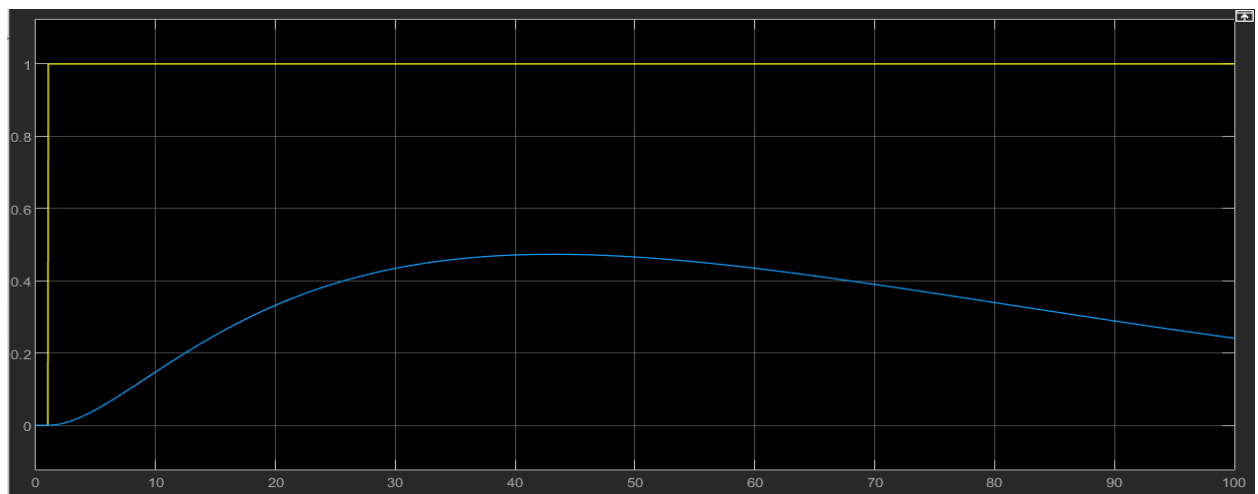




PI controller:

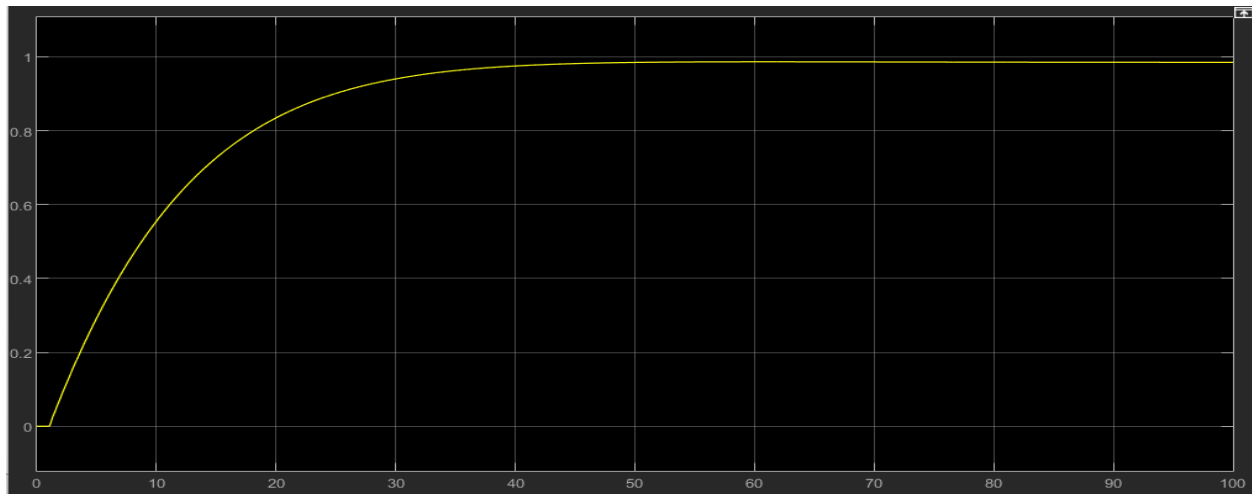


PD controller:

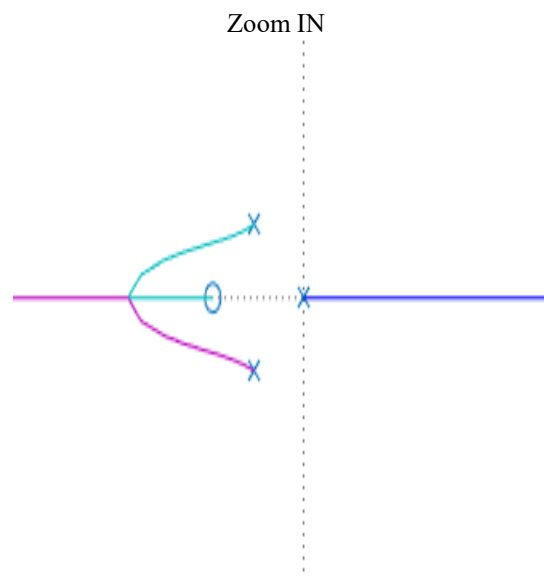
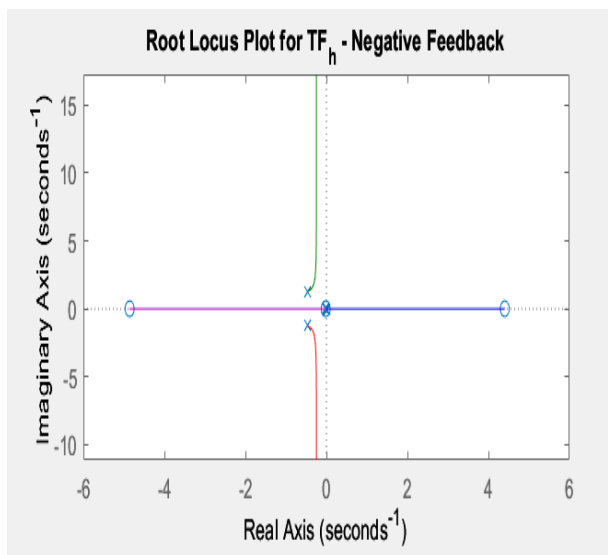


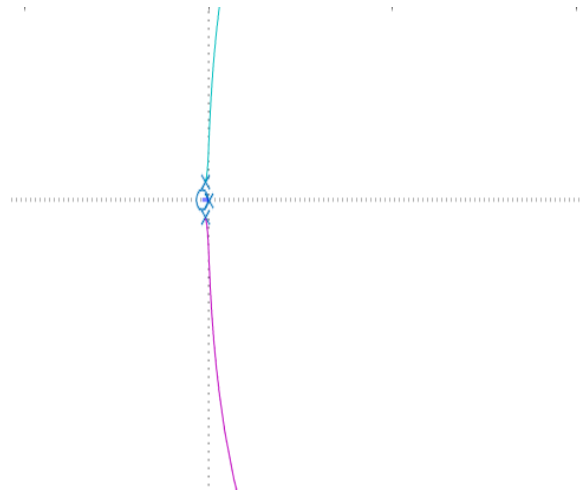
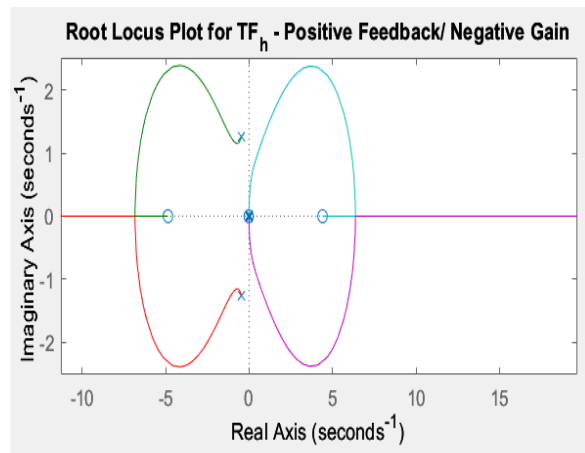
Since the open loop transfer function is of **Type 0** (that means no integrator), by no means we can get zero steady state for step input. Therefore, there is always finite error and can be reduced by choosing large gain.

PID controller:



Q4. The stable solution can't be found from the simple PID controller. This can be observed from the following root locus of the open loop transfer function when altitude only is being feedback. However, one may design lead compensator that attracts the locus to the left, but not sure!





From the root locus, one can understand that feeding the sole height/altitude is causing instability, at least for proportional controller. Of course, this was not unexpected at first. In the fixed wing aircraft, we don't have vertical actuator to hold the altitude. The absence of this actuator cause a significant challenge when we try to hold the altitude with other actuators like elevator. For example, when we use elevator to achieve the goal, due to coupling of the dynamics that are related to elevator, we will have significant challenge with the outer feedback control only. Just the analogy would be we're trying to use the elevator to do a job it's not really built for or It's like asking a chef to fix a computer – they're good at their job, just not this one. Elevators are good at adjusting the plane's angle (pitch), not so much at keeping it at a specific height. So, we need extra tricks to achieve the goal!!

Q5. With inner loop design, however, more degree of freedom to choose the parameter can be achieved. In the inner loop, three states of the plant are being feedback in determining the elevator input. And also for simplicity, elevator is assumed to be ideal (i.e. without any lag).

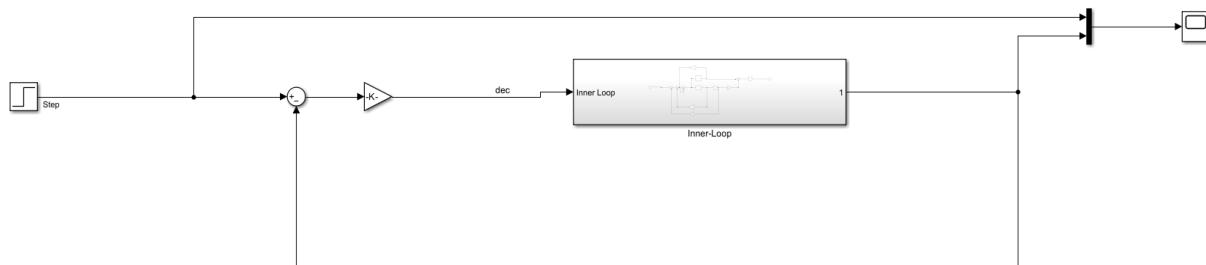


Fig. Outer loop.

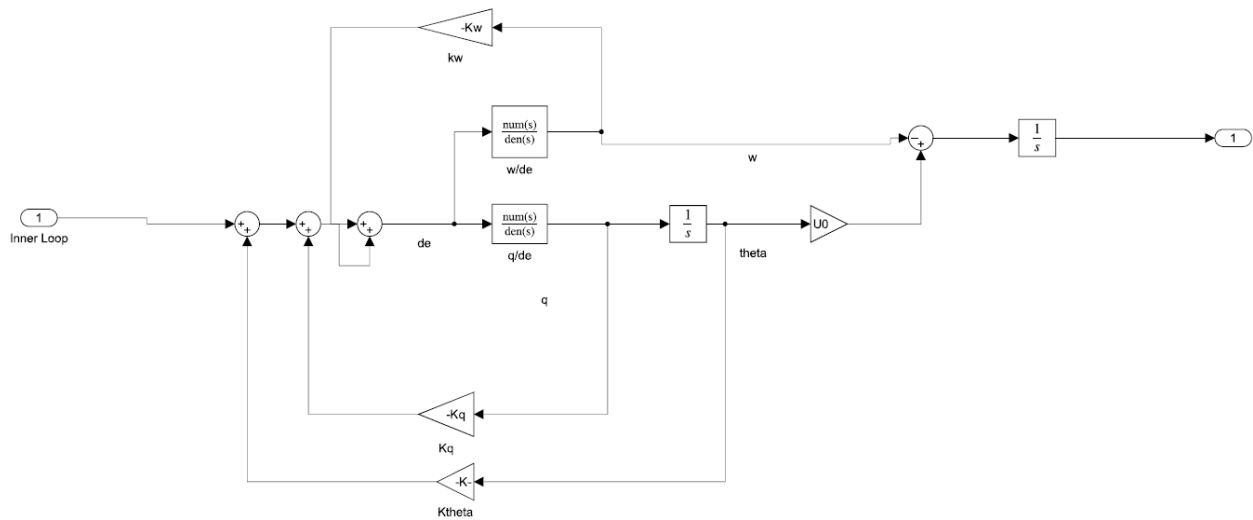


Fig. Inner loop

In order to determine the gain in the inner loop, I used controllability technique to place my pole at the desired location. And then manually tuned the outer loop. The result is very close to the one shown on the lecture material and shown below.

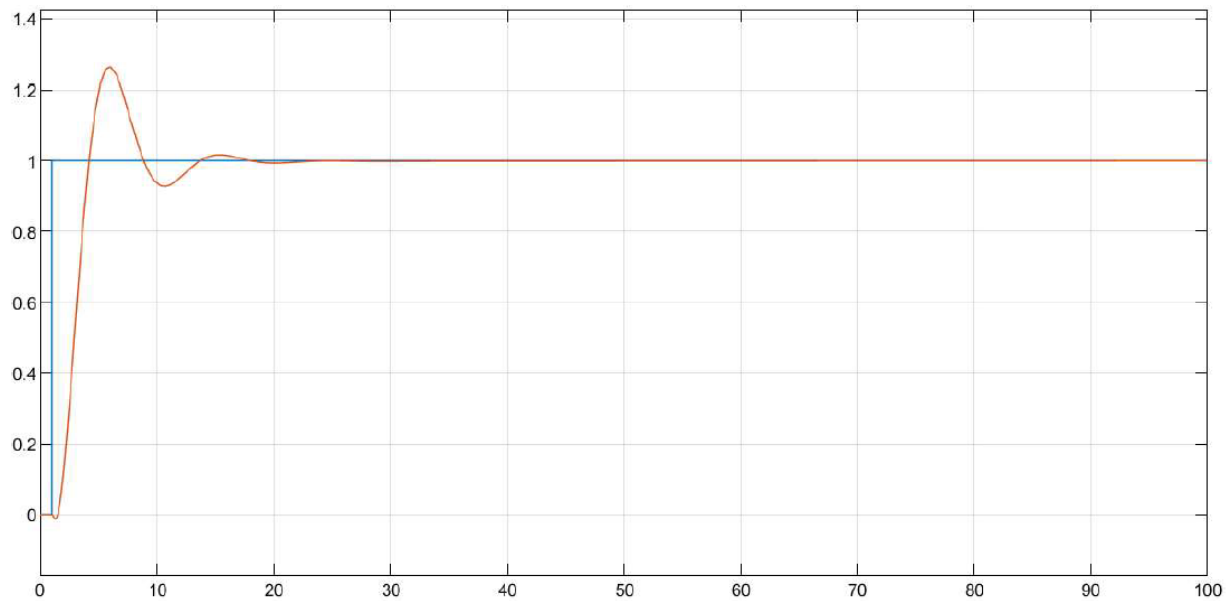


Fig. Result